$$BaO_2 + Mg \longrightarrow BaO + MgO$$

During the above reaction a large amount of heat is evolved (temperature up to 2400°C, is generated and the reaction enthalpy is : 852 kJ mol^{-1}) which facilitates the reduction of Cr_2O_3 by aluminium power.

$$Cr_2O_3 + 2Al \xrightarrow{\Delta} 2Cr + Al_2O$$

Active metals such as sodium, potassium and calcium can also be used to reduce the metal oxide

$$B_2O_3 + 6Na \longrightarrow 2B + 3Na_2O$$
$$Rb_2O_3 + 3Mg \longrightarrow 2Rb + 3MgO$$
$$TiO_2 + 2Mg \longrightarrow Ti + 2MgO$$
$$ThO_2 + 2Ca \xrightarrow{1250\ K} Th + 3CaO$$

Auto-reduction:

Simple roasting of some of the ores give the crude metal. In such cases, the use of reducing agents is not necessary. For example, mercury is obtained by roasting of its ore cinnabar (HgS)

$$HgS\ (s) + O_2\ (g) \longrightarrow Hg\ (l) + SO_2\uparrow$$

1.4 Thermodynamic principle of metallurgy

As we discussed, the extraction of metals from their oxides can be carried out by using different reducing agents. For example, consider the reduction of a metal oxide M_xO_y.

$$\frac{2}{y}M_xO_y\ (s) \longrightarrow \frac{2x}{y}M\ (s) + O_2\ (g)\ \text{------}\ (1)$$

The above reduction may be carried out with carbon. In this case, the reducing agent carbon may be oxidised to either CO or CO_2.

$$C + O_2 \longrightarrow CO_2\ (g)\ \text{------}\ (2)$$
$$2C + O_2 \longrightarrow 2CO\ (g)\ \text{------}\ (3)$$

If carbon monoxide is used as a reducing agent, it is oxidised to CO_2 as follows,

$$2CO + O_2 \longrightarrow 2CO_2\ (g)\ \text{------}\ (4)$$

A suitable reducing agent is selected based on the thermodynamic considerations. We know that for a spontaneous reaction, the change in free energy (ΔG) should be negative. Therefore, thermodynamically, the reduction of metal oxide [equation (1)] with a given

XII U1 Metallurgy - Jerald.indd 10 3/29/2019 6:33:52 PM